**Homework 27: 10.7-10.9** **Name** _____

**AP Calculus BC** **Due** _____

**Period** _____ **Score** _____ **/ 100**

## Skill Builder 10.7 – Alternating Series Tests for Convergence

Use the Alternating Series Test to determine the convergence or divergence of each of the following series.

| | |
|---|---|
| **1.)** $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+1}$ | **2.)** $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{3n-1}$ |
| **3.)** $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n-1}$ | **4.)** $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n^2}{n^2+2}$ |
| **5.)** $\sum_{n=1}^{\infty} \frac{(-1)^{n}}{\sqrt{n}}$ | **6.)** $\sum_{n=3}^{\infty} \frac{(-1)^{n} n}{\ln n}$ |

| | |
|---|---|
| **7.)** $\sum_{n=1}^{\infty} \cos(n\pi)$ | **8.)** $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n}$ |
| **9.)** $\sum_{n=1}^{\infty} (-1)^n n e^{-n}$ | |

# Skill Builder 10.8 –Ratio Test (and Root Test) for Convergence

Use the Ratio Test to determine the convergence or divergence of each of the following series.

| | |
|---|---|
| **1.)** $\sum_{n=0}^{\infty} \frac{n!}{4^n}$ | **2.)** $\sum_{n=0}^{\infty} \frac{4^n}{n!}$ |
| **3.)** $\sum_{n=1}^{\infty} n\left(\frac{3}{4}\right)^n$ | **4.)** $\sum_{n=1}^{\infty} \frac{3^n}{n^3}$ |
| **5.)** $\sum_{n=1}^{\infty} \frac{2^n}{(n+1)!}$ | **6.)** $\sum_{n=1}^{\infty} \frac{5^n}{3^n+1}$ |

Use the Root Test to determine the convergence or divergence of each of the following series.

| | |
|---|---|
| **7.)** $\sum_{n=1}^{\infty}\left(\frac{n}{3n+1}\right)^n$ | **8.)** $\sum_{n=1}^{\infty}\left(\frac{3n}{n+1}\right)^n$ |
| **9.)** $\sum_{n=0}^{\infty} e^{-n}$ | |

Test for convergence or divergence of the following using any test that we have discussed in this unit. Identify the test and show your work.

| | |
|---|---|
| **10.)** $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}4}{n}$ | **11.)** $\sum_{n=1}^{\infty}\frac{6}{n}$ |

| | |
|---|---|
| **12.)** $\sum_{n=1}^{\infty} \frac{n}{2^n + 5}$ | **13.)** $\sum_{n=1}^{\infty} \frac{7}{2\sqrt{n^3}}$ |
| **14.)** $\sum_{n=1}^{\infty} \frac{n5^n}{n!}$ | **15.)** $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$ |

# Skill Builder 10.9 – Determining Absolute or Conditional Convergence

Determine whether the series converges absolutely, or conditionally, or diverges. Be sure to show all work.

**1.)** $\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n}$

**2.)** $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$

**3.)** $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n^2}{(n+1)^2}$

**4.)** $\sum_{n=2}^{\infty} \frac{(-1)^n}{n\ln(n)}$

**5.)** $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+4}}$

**6.)** $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\sqrt{n}}$

**7.)** $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{n^2}$

# Enrichment: Topics 10.1-10.9 – Series Convergence – MCQ Practice

**1.)** Which of the following sequences converge?

**I.** $a_n = \left\{ \dfrac{2n+1}{3n-2} \right\}$ **II.** $a_n = \left\{ \dfrac{n^2-1}{n^3+1} \right\}$ **III.** $a_n = \left\{ \dfrac{\sin n}{n} \right\}$

**(A)** I only **(B)** II only **(C)** III only **(D)** II and III only **(E)** I, II, and III

**2.)** Consider the following two statements: Which words describe when they are true?

**Statement 1:** If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then the sequence $\{a_n\}$ is convergent to zero.

**Statement 2:** If the sequence $\{a_n\}$ is convergent to zero, then the series $\sum_{n=1}^{\infty} a_n$ is convegent.

**(A)** $\begin{cases} 1\text{:sometimes} \\ 2\text{:sometimes} \end{cases}$ **(B)** $\begin{cases} 1\text{:sometimes} \\ 2\text{:always} \end{cases}$ **(C)** $\begin{cases} 1\text{:always} \\ 2\text{:sometimes} \end{cases}$ **(D)** $\begin{cases} 1\text{:always} \\ 2\text{:always} \end{cases}$ **(E)** $\begin{cases} 1\text{:never} \\ 2\text{:sometimes} \end{cases}$

**3.)** If $f(x) = \sum_{n=0}^{\infty} \left( \dfrac{x}{5} \right)^n$, what is the value of $f(2) - f(-2)$?

**(A)** $\dfrac{5}{4}$ **(B)** $\dfrac{2}{5}$ **(C)** $\dfrac{20}{21}$ **(D)** $\dfrac{5}{3}$ **(E)** divergent

**4.)** For what value of $p$ does the infinite series $\sum_{n=1}^{\infty} \dfrac{\sqrt{n^3}}{n^{2p+1}}$ converge?

**I.** $p = \dfrac{5}{6}$ **II.** $p = \dfrac{3}{4}$ **III.** $p = 1$

**(A)** I only **(B)** I and II only **(C)** I and III only **(D)** II and III only **(E)** I, II, and III

**5.)** Which of the following is absolutely convergent?

**I.** $\sum_{k=0}^{\infty} \frac{(-1)^k}{k+1}$ **II.** $\sum_{k=0}^{\infty} \frac{(-1)^k}{\sqrt{k+1}}$ **III.** $\sum_{k=0}^{\infty} \frac{(-1)^k}{(k+1)^2}$

**(A)** I only **(B)** II only **(C)** III only **(D)** I and III only **(E)** I, II, and III

**6.)** When determining the possible convergence of $\sum_{n=1}^{\infty} \sin \frac{25}{n}$, the Limit Comparison Test is used, comparing the series to $\sum_{n=1}^{\infty} \frac{1}{n}$. Because the resulting limit using the comparison is _____, the conclusion is that the series $\sum_{n=1}^{\infty} \sin \frac{25}{n}$ _____.

**(A)** 1; convergent **(B)** 1; divergent **(C)** 25; convergent **(D)** 25; divergent **(E)** 0; inconclusive

**7.)** Which of the following series converge?

**I.** $\sum_{n=1}^{\infty} \frac{25^n}{n!}$ **II.** $\sum_{n=5}^{\infty} \frac{(n+1)(n-1)}{n(n+2)(n-2)(n-4)(n+1)}$ **III.** $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

**(A)** I only **(B)** II only **(C)** III only **(D)** I and II only **(E)** I, II, and III

**8.)** For what values of $p$ will $\sum_{n=1}^{\infty} \frac{5}{n^{4p+2}}$ and $\sum_{n=0}^{\infty} \left(\frac{2p+4}{5}\right)^n$ both be convergent?

**(A)** $p < -\frac{1}{4}$ and $p > \frac{1}{2}$ **(B)** $-\frac{1}{4} < p < \frac{1}{2}$ **(C)** $-\frac{1}{2} < p < \frac{1}{4}$

**(D)** $-\frac{1}{2} < p < -\frac{1}{4}$ **(E)** no values of $p$

**9.)** Which of the following series converge?

**I.** $\sum_{n=1}^{\infty} \left(\frac{4}{n} - \frac{2}{n^2}\right)^n$ **II.** $\sum_{n=1}^{\infty} \left(\frac{\ln n^{100}}{n}\right)^n$ **III.** $\sum_{n=1}^{\infty} \frac{n}{2^n}$

**(A)** I only **(B)** II only **(C)** III only **(D)** II and III only **(E)** I, II, and III

**10.)** Which of the following series converge?

**I.** $4-4+4-4+4-4+\ldots$ **II.** $4-2+1-\frac{1}{2}+\frac{1}{4}-\ldots$ **III.** $4+4+2-2+1+1+\frac{1}{2}-\frac{1}{2}+\frac{1}{4}+\frac{1}{4}+\ldots$

**(A)** I only **(B)** II only **(C)** III only **(D)** II and III only **(E)** I, II, and III

**11.)** Which of the following series converge?

**I.** $\sum_{n=0}^{\infty} \left(\frac{10e}{9\pi}\right)^n$ **II.** $\sum_{n=0}^{\infty} \left(\frac{e^n+1}{e^n+n}\right)$ **III.** $\sum_{n=2}^{\infty} \frac{1}{\ln n}$

**(A)** I only **(B)** II only **(C)** III only **(D)** I and III only **(E)** none converge

**12.)** For what values of $k$, $k > 0$, will both $\sum_{n=1}^{\infty} \frac{(-1)^{n(k+1)}}{n^{2/3}}$ and $\sum_{n=0}^{\infty} \left(\frac{k}{4}\right)^{n+1}$ both be convergent?

**(A)** all even values of $k$ **(B)** all odd values of $k$ **(C)** $k < 4$

**(D)** $k = 2$ only **(E)** there are no integer values of $k$ that will make both series converge

**13.)** The Ratio Test is applied to $\sum_{n=1}^{\infty} \frac{\pi^{n/3}}{(2n-1)!}$. What is the conclusion?

**(A)** convergent because $\lim_{n\to\infty} \frac{\sqrt[3]{\pi}}{(2n+1)!} < 1$ **(B)** divergent because $\lim_{n\to\infty} \frac{\sqrt[3]{\pi}}{4n^2+2n} < 1$

**(C)** convergent because $\lim_{n\to\infty} \frac{\sqrt[3]{\pi}}{4n^2} < 1$ **(D)** divergent because $\lim_{n\to\infty} \frac{\sqrt[3]{\pi}}{4n^2} < 1$

**(E)** convergent because $\lim_{n\to\infty} \frac{\sqrt[3]{\pi}}{4n^2+2n} < 1$

**14.)** If $a$ and $b$ are positive and $a < b$, find the value of $\sum_{n=0}^{\infty} \left(\frac{a}{b}\right)^n - \sum_{n=0}^{\infty} \left(-\frac{a}{b}\right)^n$.

**(A)** 1 **(B)** $\frac{2ab}{b^2-a^2}$ **(C)** $\frac{b^2}{b^2-a^2}$ **(D)** $\frac{a^2}{b^2-a^2}$ **(E)** divergent

**15.)** Which statement(s) are true about the series $\sum_{n=0}^{\infty}(-1)^n \frac{\sqrt{n}}{n+10}$ and what is the conclusion about the series?

**I.** $\lim_{n\to\infty} \frac{\sqrt{n}}{n+10} = 0$ **II.** $\frac{\sqrt{n+1}}{n+11} < \frac{\sqrt{n}}{n+10}$ for all values of $n$ **III.** $\frac{\sqrt{n+1}}{n+11} < \frac{\sqrt{n}}{n+10}$ for $n \geq 10$

**(A)** I only; convergent **(B)** I and II; convergent **(C)** I and III; convergent

**(D)** I only; divergent **(E)** none; divergent

**16.)** When applying the Ratio Test to the following series, which of them gives an inconclusive result?

**I.** $\sum_{n=0}^{\infty} \frac{(-1)^n}{n^2+1}$ **II.** $\sum_{n=0}^{\infty} \frac{n^2}{e^{n^2}}$ **III.** $\sum_{n=1}^{\infty} \tan\frac{1}{n}$

**(A)** I only **(B)** II only **(C)** III only **(D)** I and III only **(E)** I, II, and III

**17.)** Find the sum of the series: $\frac{1}{2}-\frac{1}{3}+\frac{1}{4}-\frac{1}{9}+\frac{1}{8}-\frac{1}{27}+\frac{1}{16}-\frac{1}{81}+\ldots$

**(A)** 1 **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{6}$ **(D)** $\frac{1}{3}$ **(E)** divergent